MIE 446 · Aerospace Structures · Homework 2B · Due Thursday, November 5, 2026 (Canvas shows the time)

Homework 2B: Integrate, Mix, and Trace the Load

Flight-control forces, moments and load paths · Parts 2, 4 and 5

Name: _____ Team: _____

The question: When a control device changes an aerodynamic force, which aircraft moments change, and where does the structure receive the load?

HW 2A read Lilienthal's measured roll and yaw coefficients and practiced $\mathbf{M} = \mathbf{r} \times \mathbf{F}$. HW 2B uses both: you integrate a prescribed force increment along a wing, calculate moments about the center of gravity (CG), mix two elevon commands under actuator limits, and follow a control load into local structure. The sequence is **predict → sketch → compute → cross-check → explain.**

How to work. Use your existing three-person wing team to discuss and independently cross-check calculations. Rotate the roles of predictor, calculator and checker. **Each student submits an individually authored report, MATLAB file and Excel workbook, plus the standard one-slide brief, README and process note.** Identify collaborators. Include the individual reflection described at the end of this handout.

Part	Task	Points
2	Integrate spanwise force and moment increments in MATLAB and Excel	35
4	Mix elevon commands for five cases, impose limits, and calculate moments	20
5	Trace structural loads and calculate local string tension	10
	HW 2B total (HW 2A carried Parts 1 and 3, 35 points)	**65**

Question points organize the technical tasks. The course homework rubric is applied to the complete HW 2 portfolio submitted with HW 2B; see the last page.

Download and open before starting

1. Open Lecture 1 in Google Colab. Use **File → Save a copy in Drive** if you plan to change its controls.
2. Download MIE446_HW2B_Starter.m and MIE446_HW2B_Excel_Starter.xlsx into one folder. Keep original copies; rename working copies with your surname.
3. Open the MATLAB file in the Editor, set the Current Folder to that folder, run the setup section, then use **Run Section** for each numbered section. In MATLAB Online, upload the file first.
4. Open the Excel workbook and save your own copy. Gray cells are intentionally blank. Enter **formulas**, not results copied from MATLAB.

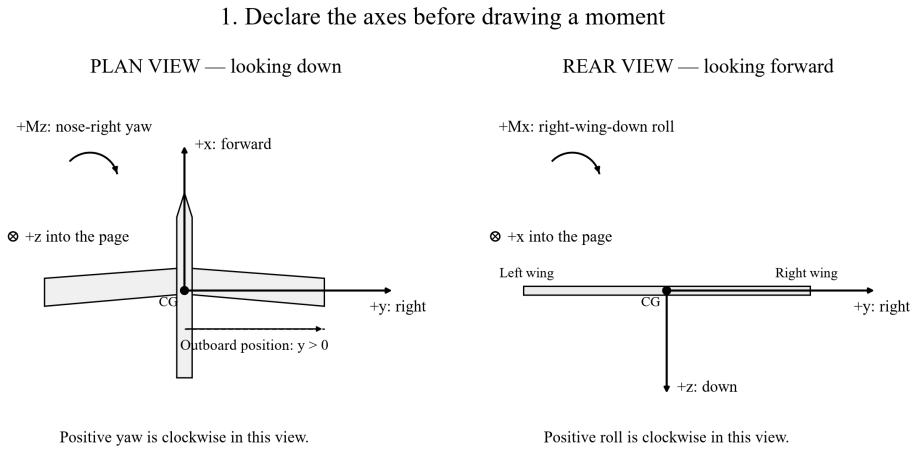
The starter uses standard MATLAB functions and no add-on toolbox. NaN means "unfinished"; a file that runs with NaN entries is not a completed submission.

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1 · Start here. Keep this handout beside the lecture notebook and your two computation files.

Coordinate axes and moment reference points

Notebook route: find **2D. Aircraft rotations**, **2E. Center of gravity**, then **7A-3b. A vertical vane is not automatically a pure-yaw device**. Use Ctrl+F (Command+F on Mac) to find the section title. Read Figure 7A-3b-1 before the equations.



Lecture 1, Figure 7A-3b-1. Aircraft right means the pilot's right looking forward. An axis drawn into the page is not absent.

Use a right-handed body-axis system: **+x forward, +y right, +z down**. Positive M_x tends to lower the right wing; positive M_y tends to raise the nose; positive M_z tends to turn the nose right. A moment tendency does not by itself predict the complete flight trajectory.

$$\mathbf{M}_{CG} = \mathbf{r} \times \mathbf{F}, \quad \mathbf{r} = (x, y, z).$$

$$M_x = yF_z - zF_y, \quad M_y = zF_x - xF_z, \quad M_z = xF_y - yF_x.$$

The coordinates in $\mathbf{r}$ run **from the CG to the force application point**. Use metres and newtons, giving N·m. A positive coordinate locates a point; it is not automatically the perpendicular lever arm of every force component.

> **Do not carry coordinate signs between different problems without checking.** HW1's airfoil coordinates had x from leading edge to trailing edge and y upward. In HW 2, Parts 2–4 use body axes and moments about the CG; Part 5 uses a local aerodynamic moment about a vane pivot. The historical paper reports its own aerodynamic-axis coefficients; retain those labels and signs in Part 1. A lowercase roll coefficient c_l is not the lift coefficient C_L.

Practice example — different numbers from the assessed cases

A force $\mathbf{F} = (0, 0, 3)$ N acts at $\mathbf{r} = (0.20, 0, 0)$ m. Then $M_x = 0$, $M_y = -(0.20)(3) = -0.60$ N·m, and $M_z = 0$. Sketch why a downward force ahead of the CG produces a nose-down tendency.

```
r_demo = [0.20, 0, 0];      % row vector, m
F_demo = [0, 0, 3];         % row vector, N
M_demo = cross(r_demo,F_demo); % returns [Mx My Mz]
```

cross(F_demo,r_demo) gives the opposite sign. Use the physical definition $\mathbf{r} \times \mathbf{F}$, not whichever order produces the expected answer.

Define the model before entering equations

Notebook route: 5. From nondimensional coefficients to forces and moment for dynamic pressure; **7A-3b** and its Figure 7A-3b-2 for signs; **7A-5. Predict the forces; audit the load path** for the physical mechanism.

Input	Value	Input	Value
Density ρ	1.225 kg/m³	Baseline speed V	15 m/s
Half-span $s_{\max}$	0.450 m	Activation station a	0.300 m
Root chord c_r	0.160 m	Tip chord c_t	0.100 m
Lift-loss factor A_z	0.30	Drag-increase factor A_x	0.10

Let s measure distance along the **right** half-wing. Use a linearly tapered chord and an activation function that grows from zero at a to one at the tip:

$$c(s) = c_r + (c_t - c_r)\frac{s}{s_{\max}}, \qquad q = \frac{1}{2}\rho V^2,$$

$$g(s) = \begin{cases} 0, & 0 \le s < a, \\ \dfrac{s-a}{s_{\max}-a}, & a \le s \le s_{\max}. \end{cases}$$

The supplied **force increments per unit span** are

$$\Delta f_z(s) = q\,c(s)A_z g(s), \qquad \Delta f_x(s) = -q\,c(s)A_x g(s).$$

They have units N/m, because Pa × m = N/m. Here $\Delta f_z > 0$ means **loss of upward lift relative to the baseline**; it does not mean the complete wing lift acts downward. The added drag points aft, so $\Delta f_x < 0$.

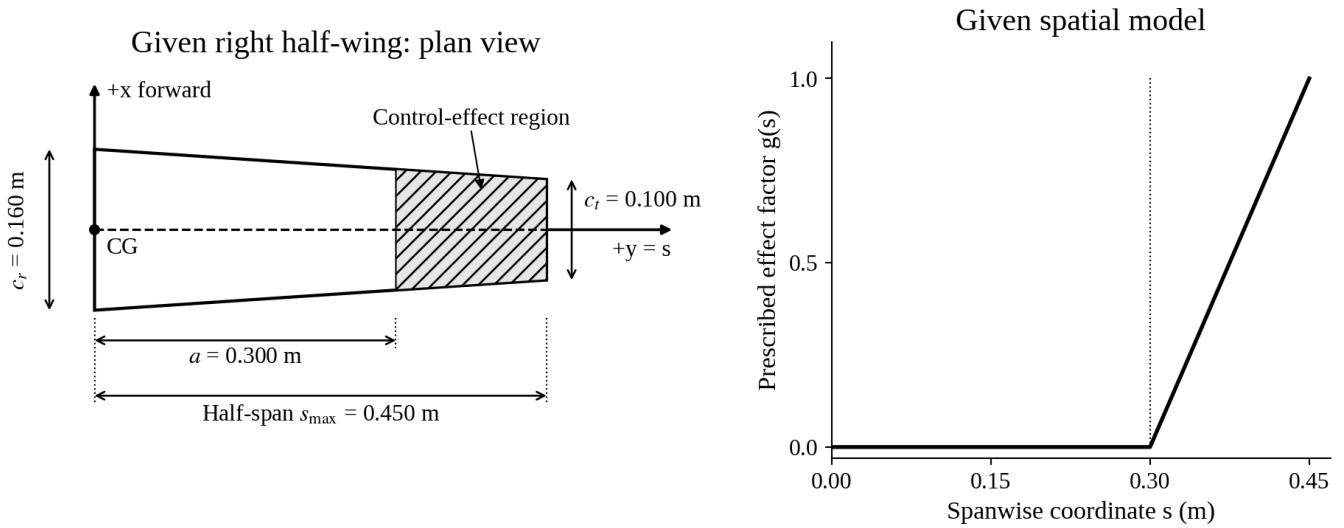
> **Model boundary.** These prescribed laws are an educational model, not measured Lilienthal data. For the moment calculation, idealize every force application point as $\mathbf{r}(s) = (0, s, 0)$ relative to the CG; Part 2A asks what this choice does to ΔM_y. The model dimensions resemble a small printed wing, but this is not a strength validation, a wing-root reaction calculation, or a prediction of its flight performance.

Excel input-cell map — WingLoads sheet

Input cell	Meaning	Input cell	Meaning
B5	ρ	B6	V
B7	$s_{\max}$	B8 / B9	Root / tip chord
B10	Activation a	B11 / B12	A_z / A_x
B13	Intervals, 90	B14	Dynamic pressure q

Part 2A · Geometry, forces, and moment integrals · 5 points

Locate the active region and the force line



Dashed force-reference line: r = (0, s, 0). This is an instructional geometry, not a flight-qualified design.

Instructional geometry. The application line $\mathbf{r} = (0, s, 0)$ is a declared force-reference idealization; it does not specify the actual aerodynamic center of a real wing.

Predict, then write the four integrals

Sketch the right half-wing and CG in plan and rear views. Mark the active region and both force increments. Predict the signs of all four results before running code:

$$\Delta F_z = \int_0^{s_{max}} \Delta f_z \, ds, \qquad \Delta F_x = \int_0^{s_{max}} \Delta f_x \, ds,$$

$$\Delta M_x = \int_0^{s_{max}} s \Delta f_z \, ds, \qquad \Delta M_z = -\int_0^{s_{max}} s \Delta f_x \, ds.$$

Explain why the force integrals have units N and the moment integrals N·m. No derivation of the cross-product formula or exact antiderivative is required.

The missing components. Use the component formulas on page 2 to show that this model gives $\Delta F_y = 0$ and $\Delta M_y = 0$ exactly, and name the idealization responsible for $\Delta M_y = 0$. Without new code, state the sign of ΔM_y if the same ΔF_z acted along a line 0.03 m **aft** of the CG, and name the tendency (nose up or nose down).

Write the prediction before calculating: which wing tends to move down, and which way does the nose tend to turn? Explain which force component creates each tendency. Then use the formulas to test your reasoning.

Optional visual check: point to the first station at which g is nonzero and compare the shapes you expect for chord and load. The activation function grows outward while the chord decreases; the load is their product, not either curve alone.

Build arrays, then integrate them

Open **Sections 2A–2C** of the starter. Replace each complete right-hand side marked TODO; leave the supplied variable name in place. Run the setup and Section 2A before Section 2B.

1. **Create the station grid.** The supplied line s = linspace(0,halfSpan,N+1).'; creates 91 stations for 90 intervals. linspace includes both endpoints; .' makes a column vector. In the Command Window, enter size(s), s(1), and s(end). Verify size 91-by-1 and endpoints 0 and 0.450 m.
2. **Calculate dynamic pressure.** Replace q = NaN using $q = \frac{1}{2}\rho V^2$. MATLAB uses ^ for a scalar power.
3. **Calculate chord at every station.** Translate $c_r + (c_t - c_r)s/s_{\max}$ into MATLAB with the supplied names rootChord, tipChord, and halfSpan. Check c(1) and c(end) against the input table.
4. **Activate only the outer region.** Use max(0,(s-activation)/(halfSpan-activation)) for g. Elementwise comparison with zero removes negative values inboard. This expression is used only on the supplied grid $0 \le s \le s_{\max}$. Check min(g), max(g), and g(s<activation).
5. **Calculate both load arrays.** Use the force laws in the Part 2 model definition. The product of two station arrays needs .*, for example c .* g. Ordinary c*g is matrix multiplication and is not the operation required here. Use ./ for elementwise division and .^ for elementwise powers if needed.
6. **Build the two moment integrands.** Fill mxDensity with s .* fz and mzDensity with -s .* fx. Their units are N; integrating over metres gives N·m.
7. **Apply the trapezoidal rule.** The syntax is trapz(stations,values). Fill Fz = trapz(s,fz);, then apply the same pattern to fx, mxDensity, and mzDensity. **Do not use trapz(fz)**: that assumes station spacing of one rather than 0.005 m.
8. **Run Section 2C.** When your values are finite, it displays the result table, saves station data using table and writetable, and draws three plots using subplot and plot. Read each axis label and explain the zero-load region. Open the output folder printed by the setup.

A small completed example of trapz

```
s_demo = [0; 0.2; 0.4];  % m
f_demo = [0; 3; 6];      % N/m
F_demo = trapz(s_demo,f_demo);
```

The two trapezoids are $0.2(0+3)/2 = 0.3$ N and $0.2(3+6)/2 = 0.9$ N, so MATLAB returns 1.2 N. Your assessed calculation uses the same procedure with 90 intervals.

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5 · Inspect the arrays and units before trusting the final four numbers.

Check sensitivity to grid spacing and speed

2C-1. Refine the grid [5 points]

1. Keep the baseline $V = 15$ m/s. In starter **Section 2D**, sFine contains 181 stations for $N = 180$ intervals.
2. Recompute cFine, gFine, fzFine, and fxFine on sFine. Copy the physical laws, then replace the coarse variable names with their fine-grid counterparts. Do not combine 181 station locations with 91 force values.
3. Store the four fine-grid integrals in a row in this order: [DeltaFz, DeltaFx, DeltaMx, DeltaMz]. In the starter this row is named fine; the corresponding coarse row is baseline.
4. For each quantity Q, calculate

$$\text{relative change (percent)} = 100\frac{|Q_{180} - Q_{90}|}{|Q_{180}|}.$$

Use 100*abs(fine-baseline)./abs(fine). All four reference magnitudes are nonzero for this model. A change below **0.1%** is adequate here. State whether each result passes. Grid agreement checks numerical resolution; it does not validate the aerodynamic model.

Resolution test. Where does the activation corner $s = a = 0.300$ m fall relative to the $N = 90$ and $N = 180$ stations? Set N = 10 in Section 2A (stations every 0.045 m) and run Sections 2A–2D once; the convergence table then compares $N = 10$ with $N = 180$ and is saved as Q2_convergence_N10.csv. Explain why the change is now much larger, then restore N = 90 and rerun. A check that cannot fail tells you little.

2C-2. Predict and test speed scaling [5 points]

1. Before running starter **Section 2E**, predict the ratio of every force and moment at 10 and 20 m/s relative to 15 m/s. Write your prediction as $(V/15)^2$ and evaluate it.
2. The supplied loop is for j = 1:numel(speeds), where speeds = [10;15;20]. At each iteration, use speeds(j) to calculate a new dynamic pressure qj.
3. Recompute fzj and fxj using qj and the coarse $N = 90$ geometry. Compute all four integrals again. Store the row in speedResults(j,:); the colon means “all four columns.”
4. Make a table of speed and the four results. Divide each 10 and 20 m/s result by the matching 15 m/s result. Compare with the ratios predicted in Step 1. Keep geometry and coefficients fixed during this experiment.
5. Explain in two sentences why the results scale with V^2 **within this model** and which assumption could make that scaling fail for a real control device.

Checks you can make without an answer key

Inboard of 0.300 m both load increments should be zero. Changing speed must not change the chord or activation curves. The resultant forces should retain their predicted signs. Because the two load laws share the same spatial shape, the factor ratio A_z/A_x also constrains the ratios $\Delta F_z/(-\Delta F_x)$ and $\Delta M_x/\Delta M_z$; calculate and explain this check. Finally, $\Delta M_x/\Delta F_z$ is the load-weighted mean station of the lift-loss increment, so it must lie between 0.300 and 0.450 m.

Submit from MATLAB: labeled load plots; the four baseline results with units; a four-row $N = 90$ versus $N = 180$ convergence table with the $N = 10$ resolution test; and the speed table with predicted and computed ratios. Include your completed script, not screenshots of code alone.

6 · Change one modeling input at a time and explain what the comparison actually tests.

Part 2D · Independent Excel integration · 10 of the 35 Part 2 points

Build the trapezoids in the WingLoads sheet

Use the supplied **WingLoads** sheet. Inputs are in B5:B13; B14 is for dynamic pressure. Keep $V = 15$ m/s and $N = 90$ for the submitted baseline. There are 91 station rows, **21:111**; A is station index, B is s in metres. A/B are already filled.

Use the Excel input-cell map in the Part 2 model definition (page 3); fixed input references use dollar signs.

1. Click **B14** and type =0.5*B5*B6^2. Press Enter. Confirm the result is positive and has units Pa.
2. Enter the following formulas in row 21, then select **C21:H111** and use **Home → Fill → Down** (or Ctrl+D). Absolute references such as $B$8 remain fixed; B21 changes with the station row.

Cell	Quantity	Formula
C21	Chord	=$B$8+($B$9-$B$8)*B21/$B$7
D21	Activation	=MAX(0,(B21-$B$10)/($B$7-$B$10))
E21	Δf_z	=$B$14*C21*$B$11*D21
F21	Δf_x	=-$B$14*C21*$B$12*D21
G21	Roll integrand	=B21*E21
H21	Yaw integrand	=-B21*F21

3. Rows represent **stations**; a trapezoid spans two adjacent stations. Leave the supplied “n.a.” entries in I21:L21 unchanged because row 21 has no preceding interval. Enter the first interval’s formulas in row 22:

Cell	Trapezoid contribution	Formula
I22	ΔF_z	=(B22-B21)*(E22+E21)/2
J22	ΔF_x	=(B22-B21)*(F22+F21)/2
K22	ΔM_x	=(B22-B21)*(G22+G21)/2
L22	ΔM_z	=(B22-B21)*(H22+H21)/2

4. Fill **I22:L111** down. Sum I22:I111 in **E5**, J22:J111 in **E6**, K22:K111 in **E7**, and L22:L111 in **E8**. Example: =SUM(I22:I111). Include all 90 intervals.
5. Make a four-row comparison table in your report: quantity, MATLAB $N = 90$, Excel, absolute difference. Agreement within 10^{-9} N for forces and 10^{-9} N·m for moments is adequate; retain full precision during computation. Copy the full-precision MATLAB baseline printed by the script (or the exported Q2_baseline_totals.csv) into F5:F8. Enter =E5-F5 in G5 and fill through G8. Report ABS(G5) and its counterparts as absolute differences. Do not copy rounded display values.
6. Create one Excel **XY Scatter with lines** chart using B21:B111 as numerical x and E21:E111 as y. Label axes with units. Do not use a category-axis Line chart. Save the workbook with formulas visible when a cell is selected.

If your Excel uses semicolons between function arguments, replace the comma in MAX accordingly. Changing B13 alone does not create extra station rows; grid refinement is assessed in MATLAB.

Part 4 · Elevon mixing and control limits · 20 points

Two surfaces share pitch and roll commands

Notebook route: **7A-11. What does each modern effector actually move or change?** Read the generic elevon mixer, then try the following **"7A-11. Elevon mixer — predict before running"** code cell. Use it to understand commands; the extra force model below is supplied for HW2.

Positive local deflection means **trailing edge up** on each side. Let P request nose-up pitch and R request right-wing-down roll, in this instructional configuration:

$$\delta_L = P - R, \qquad \delta_R = P + R, \qquad -15^\circ \le \delta_{L,R,\text{limited}} \le 15^\circ.$$

Case	P (deg)	R (deg)	What to look for
1	8	0	Common-mode command
2	0	5	Differential command
3	6	4	Combined command
4	10	7	A command may exceed the actuator limit
5	−6	10	Nose-down pitch with strong roll: check the lower limit

4A. Predict surface motion [4 points]

Sketch left and right trailing-edge motion for each case **before running the notebook or MATLAB**. Calculate requested δ_L, δ_R. Mark any value outside ±15°. A negative deflection means trailing edge down. These are surface commands, not aircraft pitch or bank angles.

4B. Convert commands to force increments [8 points]

For this exercise only, use the linear gain $k = 0.05$ N/deg and application points:

$$\mathbf{r}_L = (-0.15, -0.35, 0)\ \text{m}, \qquad \mathbf{r}_R = (-0.15, +0.35, 0)\ \text{m},$$
$$\Delta\mathbf{F}_L = (0, 0, k\delta_L), \qquad \Delta\mathbf{F}_R = (0, 0, k\delta_R).$$

Evaluate these force vectors twice: first using the requested deflections and then using the limited deflections. Positive ΔF_z is a downward force increment. Because both force application points lie aft of the CG, the signs are consistent with the teaching mixer.

$$\Delta\mathbf{M}_{\text{total}} = \mathbf{r}_L \times \Delta\mathbf{F}_L + \mathbf{r}_R \times \Delta\mathbf{F}_R.$$

Find all three components; do not omit M_z simply because you expect it to vanish. Show one complete hand calculation for **Case 3 before limiting** and check all five cases with MATLAB and Excel.

> **What this model includes.** A fixed linear force-per-degree gain, two fixed force application points, and symmetric actuator limits. It omits changes in drag, nonlinear aerodynamics, and aircraft motion. It is not the flight-control law of any named UAV, and a zero yaw result here does not establish zero yaw coupling for real elevons.

8 · First calculate local surface motion; then calculate its forces and moments.

Apply the limit to each surface separately

MATLAB: complete Sections 4A and 4B

1. The rows of commands are the five cases. In the supplied loop, P and R are already read from the current row. Set requested(j,:) to [P-R, P+R].
2. Use min(max(requested(j,:),-limit),limit) to obtain the two limited deflections. The inner max prevents a value below −15; the outer min prevents a value above +15. Test the expression on [-18, 7, 21] separately so you understand the clipping.
3. Create row force vectors [0,0,k*delta] for each side, using the matching requested or limited deflection. Compute cross(rLeft,FLeft) plus cross(rRight,FRight). Store the requested and limited moment rows separately.
4. Recover the two attained command channels using the formulas below. The starter then exports a table and draws grouped bar charts for Cases 4 and 5 using bar(...,'grouped'). In each chart, the two groups are the two surfaces.

$$P_{\text{attained}} = \frac{\delta_{L,\text{limited}} + \delta_{R,\text{limited}}}{2}, \qquad R_{\text{attained}} = \frac{\delta_{R,\text{limited}} - \delta_{L,\text{limited}}}{2}.$$

Read the output: requested and limited angles are in degrees; force increments are in newtons; moments are in N·m. The attained P/R values are mixed commands, not measured attitude angles. Compare Cases 4 and 5 before and after clipping without changing any geometry or gain.

Compare requested and attainable moments

Excel: Controls sheet, rows 22:26 and 45:49

B14 is the limit, B15 the gain, B16:B18 the left x/y/z coordinates, and E16:E18 the right x/y/z coordinates. B/C in each case row contain P/R. For the main table, enter row 22 formulas and fill through row 26. For the comparison table (header in row 44), enter row 45 formulas referencing Case 1 in row 22, then fill through row 49:

Cell	Meaning	Formula
D22 / E22	Requested left / right	=B22-C22 / =B22+C22
F22 / G22	Limited left / right	=MIN(MAX(D22,-$B$14),$B$14) =MIN(MAX(E22,-$B$14),$B$14)
H22 / I22	Limited $\Delta F_{z,L/R}$	=$B$15*F22 / =$B$15*G22
J22	Limited total M_x	=$B$17*H22+$E$17*I22
K22	Limited total M_y	=-$B$16*H22-$E$16*I22
L22	Limited total M_z	Enter the formula implied by the force vectors.
B45	Requested total M_x	=$B$17*$B$15*D22+$E$17*$B$15*E22
C45	Requested total M_y	=-$B$16*$B$15*D22-$E$16*$B$15*E22
D45	Requested total M_z	Enter the formula implied by the force vectors.
E45 / F45	Attained P / R	=(F22+G22)/2 / =(G22-F22)/2

4C. Explain saturation [8 points]

For Cases 4 and 5, compare requested versus limited deflections, attained P/R channels, and M_x/M_y. Explain why limiting one physical surface changes **both** command channels, and why the attained channels move toward zero in both cases even though different surfaces reach different limits. Which cases should remain unchanged by the limit? Explain why the attained channels are not measured aircraft attitude angles. Submit your prediction sketches, one hand calculation, result table, Case 4–5 charts, and explanation.

10 · A surface limit constrains combinations of commands, not just one abstract channel.

Part 5 · Structural load paths and local mechanism load · 10 points

Follow the reactions into the airframe

Notebook route: 7A-1. Wing structure, **7A-5. Predict the forces; audit the load path**, and **7A-14. Bring the control mechanism back to Aerospace Structures**. Use the wing-anatomy figures to identify the physical members receiving each reaction.

5A. Draw two load-path sketches [5 points]

Draw one **modern upper-surface spoiler** and one **vertically pivoting wing-tip vane**. On each sketch:

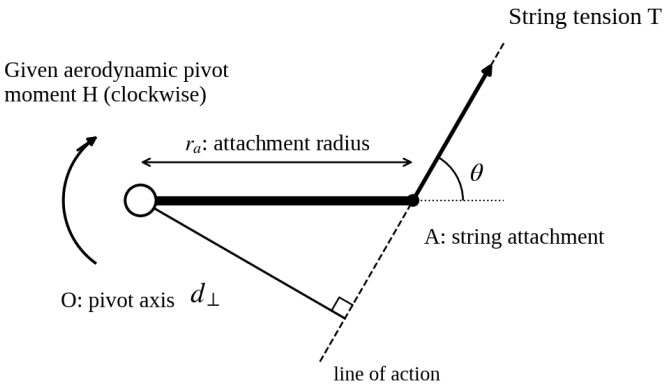
1. Identify the moving part, its hinge or pivot, and the nearby wing structure. Show an aerodynamic force arrow acting on the moving part.
2. Draw the load route from the moving part through its hinge/bracket or pivot/post into local structure, then toward the wing root.
3. Add the **separate actuation branch**: actuator and anchor for the spoiler; string/control rod and its support for the vane. Label the reaction where that branch attaches to the structure.
4. Circle one connection that could require local reinforcement. State what additional information you would need to calculate its stress, such as geometry, material properties, or reaction distribution.

Use arrows and labels, not an unannotated aircraft photo. Avoid sending all load through the actuator alone or through the hinge alone without drawing the other branch.

Sketch planning space. Lay out the aerodynamic-load and actuator-reaction branches for both mechanisms. Label the local structure receiving each branch.

Use the perpendicular moment arm

Local pivot, string force and perpendicular arm



Two branches carry the control loads

Aerodynamic load → vane or panel → pivot / hinge bracket

String or actuator force → linkage → anchor

Both branches end in supporting wing structure

Static balance about O: $T d_\perp = H,\ d_\perp = r_a \sin\theta$

Ideal, frictionless mechanism in static balance. H is a given local moment about the vane pivot, not the aircraft roll moment about the CG.

5B. Calculate string tension from a given local moment [3 points]

The aerodynamic load on a simple vane produces a given **local** moment $\boldsymbol{H = 0.12}$ N·m about the vane's pivot. The control string must balance it with an equal and opposite moment, acting through its perpendicular moment arm $\boldsymbol{d_\perp}$. Assume static moment balance and neglect friction:

$$\boldsymbol{T = \frac{H}{d_\perp}}, \qquad \boldsymbol{d_\perp = 0.010,\ 0.020,\ 0.030\ \text{m}.}$$

1. Sketch the pivot, string-force line, and the **perpendicular distance** between them. The perpendicular distance is not necessarily the attachment radius.
2. In MATLAB starter **Section 5**, replace the TODO with H ./ d. Explain why elementwise division gives one tension for each arm.
3. *Optional Excel check:* in **Controls**, H is in **B30** and the arms are in **B34:B36**; =$B$30/B34 in C34, filled through C36, should match MATLAB. Report tension in newtons.
4. Explain what happens to tension when the perpendicular arm is doubled, keeping H fixed.
5. **By hand only:** the string is attached at $\boldsymbol{r_a = 0.030}$ m from the pivot but pulls at $\boldsymbol{\theta = 30°}$ to the lever line, as in the figure. Find $\boldsymbol{d_\perp}$ and $\boldsymbol{T}$. Compare with the 0.030 m row of your table and explain the difference.

5C. Distinguish two moments [2 points]

Can you take the aircraft's global rolling moment from Part 2 and automatically use it as $\boldsymbol{H}$ in the string-tension equation? Explain using the different reference axes and free bodies. Name one additional measurement or geometric detail needed to connect a device's aerodynamic load to its local pivot moment.

Bridge to your printed wing: identify one location where a moving device would introduce a new fitting or reinforcement requirement. A labeled conceptual sketch is sufficient; this homework does not add a moving-control mechanism to the scheduled fabrication deliverables.

Make your result easy to reproduce

Submit your individually authored **Surname_HW2B_Report.pdf**, **Surname_HW2B.m**, and **Surname_HW2B.xlsx**, together with the course-required **Claim–Evidence–Check slide (.pptx and .pdf)** and a short **README**. Submit through Canvas; keep graded work out of the public repository. Include the plots and key tables in the PDF so a reader can follow your reasoning without opening the software. Keep full computation precision; round only displayed results to a sensible number of significant figures.

Part	Evidence required in your report	Points
2	Predictions and sketches; ΔM_y explanation; baseline plots/results; grid-refinement table and $N = 10$ resolution test; speed-scaling check; MATLAB–Excel comparison	35
4	Surface-motion predictions; one hand calculation; five-case requested/limited force–moment results; saturation charts and explanation	20
5	Two structural load paths; local moment sketch, tension table and inclined-string case; global/local moment distinction	10

Individual reflection and team cross-check record

Add a short paragraph naming your collaborators and recording one cross-check you performed independently. Then, in about **150–200 words**, explain (i) a force/moment sign you initially found difficult, (ii) how a sketch or independent calculation resolved it, and (iii) one result that depends on a simplifying model. This reflection supports the reasoning marks above; it is not an extra calculation problem or a separate team report.

Before submitting

1. Save, close, and reopen your MATLAB file. Confirm N = 90 in Section 2A and run it from the top. Confirm that no assessed variable remains NaN, all result tables appear, and files are saved in hw2b_outputs next to the script.
2. Open the Excel workbook and click representative cells from each part. Confirm that they contain formulas referencing the intended inputs. Verify the baseline input speed is restored to 15 m/s.
3. Check that every plot has a descriptive title or caption and labeled axes with units. Check that every moment table states its reference point and coordinate convention.
4. Read every prediction beside its computed result. If your prediction was wrong, explain the correction; do not silently remove the original reasoning.
5. Ensure you distinguish historical measurements, figure-reading estimates, and the supplied educational force laws. Include source references for copied figures.
6. If HW 2A feedback identified a sign or convention error, confirm that HW 2B does not repeat it and record the correction in your process note.

Finish the standard homework portfolio

HW 2A (35 points) and HW 2B (65 points) together form Homework 2. The question points organize the technical tasks; the established course rubric is applied once, to the complete HW 2 portfolio submitted here: **engineering judgment 40%; mechanics and numerical correctness 25%; independent verification 20%; one-slide communication 10%; reproducibility, attribution and AI disclosure 5%**.

Engineering Judgment Record: use Part 2

Designate a short subsection of your report as **“Part 2 — Engineering Judgment Record.”** Reuse the work you already completed: state the system, prescribed loads, assumed application line, prediction, model assumptions and limitations; show at least two independent checks. Suitable checks: dimensional consistency, the MATLAB–Excel comparison, a limiting case, speed scaling, or an exact hand integral of the load law. Distinguish numerical agreement from validation of the force law.

Model choice. Answer briefly, without new code: (i) Which mechanism in your HW 2A Part 1 readings shares the roll–yaw sign pattern of your Part 2 results? (ii) Which sign of A_x would reproduce the other mechanism’s pattern, and what would happen to ΔM_z? (iii) Name one feature of Lilienthal’s vane that this load law cannot represent.

One Claim–Evidence–Check slide

Create one slide in PowerPoint and export the same slide as PDF. Use one result from HW 2A or HW 2B. Include a clear claim, a readable figure or numerical result supporting it, an independent check, your confidence, and the principal limitation. Synthesize the reasoning rather than pasting a collage of screenshots. Follow the existing Canvas rotation for oral checks; this handout does not assign an additional presentation.

README and process / AI-use note

In a short README text file, state your MATLAB version, Excel version, filenames, how to run the script, and where the outputs appear. Add a process and AI-use note to the report or README: tools used, their contributions, changes you made, errors found, and how you verified the result. If applicable, write **“No generative AI used.”** Identify collaborators and sources.

If something goes wrong

Symptom	First check
MATLAB matrix-size error	Use column station arrays; inspect size; use .* for stationwise products.
Force is off by a factor related to grid spacing	Pass both arguments to trapz(s,f); include interval width in every Excel trapezoid.
Signs disagree between methods	Use cross(r,F); check +z down and the minus sign in the yaw integrand.
Excel formulas change input references when filled	Use dollar signs for fixed cells; allow station row references to change.
Excel matches MATLAB but units are wrong	Numerical agreement is insufficient: distinguish N/m, N, and N·m.
Only Cases 4–5 give unexpected coupling	Inspect each surface’s requested and limited angle before recomputing force and moment.

MATLAB function references: trapz, cross, linspace, and exportgraphics (optional figure export in R2020a or later). The supplied starter uses print to save its PNG plots.